

SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Short Answer Test

COURSE NAME: COMPUTER VISION COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)

Assessment Tool Weightage: 40%

QUESTIONS

1. A satellite image shows loss of terrain details. Write a Python program to analyze image resolution.
2. A biometric system captures face images. Write a Python program for face recognition.
3. An image appears pixelated when zoomed. Write a Python program to demonstrate image pixelation.

1. A satellite image shows loss of terrain details. Write a Python program to analyze image resolution.

Program:

```
File Edit Format Run Options Window Help
import cv2

# Read the image
image = cv2.imread(r"C:\Users\lenin\OneDrive\Pictures\Screenshots\Satellite terrain.png")

# Get image resolution
height, width, channels = image.shape

print("Image Width :", width)
print("Image Height:", height)
print("Channels      :", channels)
print("Resolution   :", width * height, "pixels")

# Display image
cv2.imshow(r"C:\Users\lenin\OneDrive\Pictures\Screenshots\Satellite terrain.png", image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:

```
Image Width : 285
Image Height: 236
Channels      : 3
Resolution   : 67260 pixels
```



2. A biometric system captures face images. Write a Python program for face recognition.

Program:

```
import cv2

# Give the full path of your image here
image_path = r"C:\Users\lenin\OneDrive\Pictures\Screenshots\kk.jpg"
# Load the Haar Cascade classifier
face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades + "haarcascade_frontalface_default.xml"
)

# Read the image
image = cv2.imread(image_path)

# Check if image is loaded
if image is None:
    print("Error: Image not found!")
    input("Press Enter to exit...")
    exit()

# Convert image to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Detect faces
faces = face_cascade.detectMultiScale(
    gray,
    scaleFactor=1.1,
    minNeighbors=5,
    minSize=(30, 30)
)

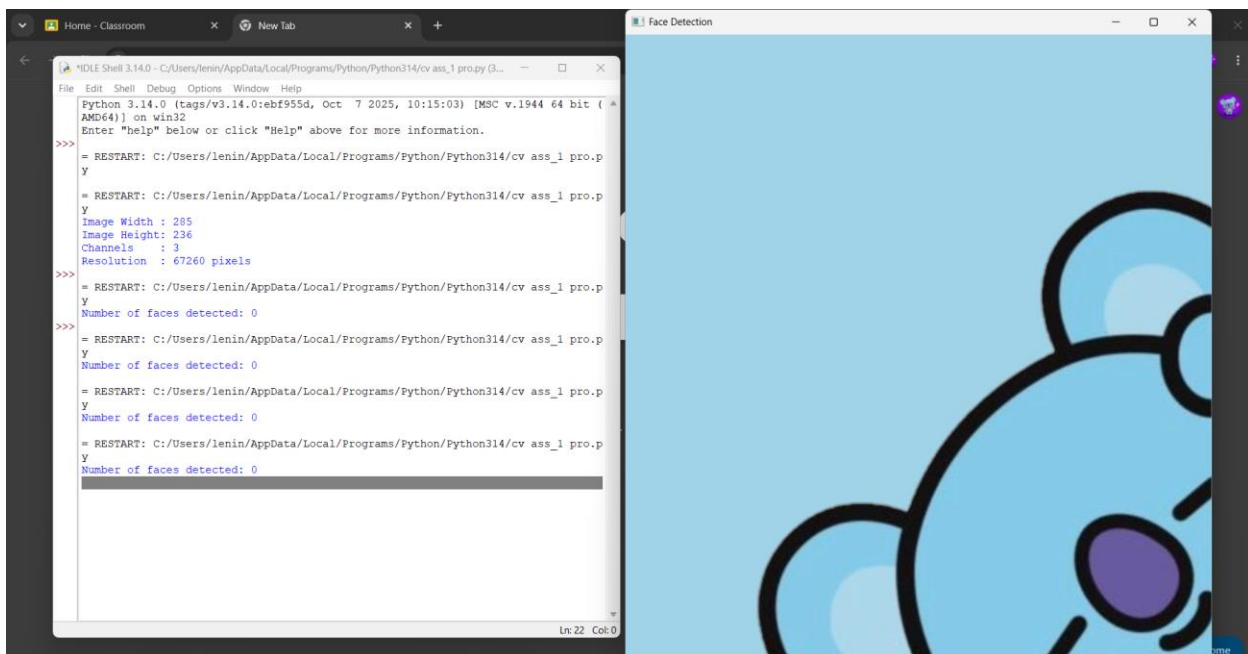
# Draw rectangles around detected faces
for (x, y, w, h) in faces:
    cv2.rectangle(image, (x, y), (x + w, y + h), (0, 255, 0), 2)

# Print number of faces
print("Number of faces detected:", len(faces))

# Show output image
cv2.imshow("Face Detection", image)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:



3. An image appears pixelated when zoomed. Write a Python program to demonstrate image pixelation.

Program:

```
File Edit Format Run Options Window Help
import cv2

# Give the full image path here
image_path = r"C:\Users\lenin\OneDrive\Pictures\Screenshots\pixelated.png"

# Read the image
image = cv2.imread(image_path)

# Check if image is loaded
if image is None:
    print("Error: Image not found!")
    input("Press Enter to exit...")
    exit()

# Resize to a very small size
small = cv2.resize(image, (50, 50), interpolation=cv2.INTER_LINEAR)

# Enlarge it back to original size using nearest-neighbor interpolation
pixelated = cv2.resize(
    small,
    (image.shape[1], image.shape[0]),
    interpolation=cv2.INTER_NEAREST
)

# Display the images
cv2.imshow("Original Image", image)
cv2.imshow("Pixelated Image", pixelated)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:

